

CASABLANCA NOUASSEUR, MoroccoWMO: **601560**Lat: **33.367N**Lon: **7.590W**Elev: **656**StdP: **14.35**Time Zone: **1.00 (E01)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	37.3	39.5	31.8	26.8	44.9	35.0	30.6	46.5	27.1	56.2	22.8	56.8	4.1	160	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.0	96.8	70.9	92.1	70.5	88.3	69.7	74.4	87.8	73.2	85.5	72.1	83.2	11.7	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.4	118.6	77.3	69.9	112.8	76.5	68.5	107.3	75.5	38.3	88.0	37.1	85.0	36.1	83.3	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.9	18.9	16.8	DB	33.5	106.6	3.0	5.1	31.4	110.3	29.6	113.2	28.0	116.1	25.8	119.8
			WB	32.0	77.7	3.2	2.3	29.7	79.3	27.8	80.7	26.1	82.0	23.7	83.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	65.2	53.5	55.0	59.1	61.8	67.0	72.4	75.9	77.0	73.4	69.4	61.2	56.1	(6)
(7)		DBStd	9.36	4.04	4.61	5.13	4.76	5.56	4.89	4.63	4.14	4.07	4.78	5.56	4.18	(7)
(8)		HDD50	25	12	9	1	0	0	0	0	0	0	0	0	3	(8)
(9)		HDD65	1427	357	281	196	120	39	1	0	0	0	14	143	276	(9)
(10)		CDD50	5578	120	148	284	353	527	673	802	838	703	601	336	193	(10)
(11)		CDD65	1504	0	1	13	23	101	224	337	373	253	150	28	1	(11)
(12)		CDH74	11532	8	25	203	288	913	1712	2648	2894	1585	1023	214	19	(12)
(13)		CDH80	4394	0	2	56	95	374	670	1109	1195	506	338	48	1	(13)
(14)	Wind	WSAvg	8.1	7.0	7.5	8.1	8.5	9.0	9.0	9.3	8.8	7.9	7.7	7.6	7.4	(14)
(15)	Precipitation	PrecAvg	14.2	2.2	1.7	1.5	1.3	0.6	0.1	0.0	0.4	1.4	2.7	2.3		(15)
(16)		PrecMax	33.1	10.2	5.4	4.0	3.6	2.5	0.6	0.3	0.2	2.4	3.9	9.1	11.9	(16)
(17)		PrecMin	7.8	0.1	0.1	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.2	0.1	0.0	(17)
(18)		PrecStd	5.4	2.2	1.4	1.0	0.9	0.6	0.1	0.1	0.0	0.6	1.1	2.2	2.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.7	77.5	86.4	89.4	98.4	100.3	103.8	102.3	96.5	93.4	85.8	76.8	(19)
(20)			MCWB	59.7	60.3	63.5	63.3	66.8	70.1	72.6	72.4	71.0	66.5	63.7	58.5	(20)
(21)		2%	DB	69.2	71.9	79.3	82.0	89.3	92.4	95.3	94.7	89.7	87.8	79.8	71.9	(21)
(22)			MCWB	58.0	59.5	61.7	64.3	67.1	70.0	71.6	72.6	71.0	66.2	62.9	58.6	(22)
(23)		5%	DB	65.9	68.0	74.1	76.3	82.8	87.4	89.9	89.7	85.2	83.6	75.3	68.2	(23)
(24)			MCWB	57.6	58.8	61.4	63.4	65.4	69.3	71.0	71.8	70.1	66.8	62.6	58.9	(24)
(25)		10%	DB	62.8	64.6	69.9	71.8	78.1	82.6	85.8	86.1	82.1	79.4	71.6	65.4	(25)
(26)			MCWB	56.5	57.4	60.2	61.4	64.7	68.3	70.5	71.3	69.3	66.7	62.1	58.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.8	64.2	66.8	69.0	71.1	73.7	76.4	76.4	74.7	71.8	69.0	64.1	(27)
(28)			MCD B	69.1	72.8	78.4	81.6	88.9	90.8	93.1	90.6	86.4	82.9	76.7	69.5	(28)
(29)		2%	WB	60.2	61.3	64.1	65.5	68.5	71.6	74.0	74.6	72.9	70.0	66.3	61.9	(29)
(30)			MCD B	65.8	68.1	74.6	77.1	83.5	86.0	88.4	87.6	82.9	79.5	73.6	66.9	(30)
(31)		5%	WB	58.5	59.6	62.3	63.6	67.0	70.2	72.6	73.4	71.6	68.8	64.5	60.4	(31)
(32)			MCD B	63.7	66.0	71.7	73.8	80.2	83.4	85.5	85.3	80.4	77.4	71.4	65.3	(32)
(33)		10%	WB	57.0	58.1	60.6	62.0	65.3	68.9	71.2	72.3	70.3	67.6	62.8	59.0	(33)
(34)			MCD B	62.0	63.8	68.5	70.8	76.1	80.4	82.5	82.8	78.9	75.4	69.7	64.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	19.8	20.3	21.3	20.2	20.4	19.4	18.9	19.0	18.9	20.7	20.0	19.5	(35)
(36)			MCDBR	24.6	25.1	29.0	28.0	29.7	27.8	26.6	24.9	23.7	27.2	25.2	24.4	(36)
(37)			MCWBR	16.0	15.5	15.7	13.9	12.1	10.1	9.0	8.3	9.0	12.1	13.4	14.9	(37)
(38)		5% WB	MCD BR	20.8	21.9	24.4	23.8	24.9	23.0	22.8	21.7	20.4	21.7	20.1	18.5	(38)
(39)			MCWBR	14.6	14.4	14.4	12.7	11.0	9.4	8.8	8.1	8.5	10.8	11.6	12.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.326	0.354	0.407	0.404	0.429	0.462	0.506	0.500	0.441	0.401	0.361	0.335	(40)	
(41)		taud	2.493	2.398	2.224	2.264	2.208	2.114	1.976	2.014	2.211	2.340	2.441	2.496	(41)	
(42)		Ebn at Noon	282	284	275	281	273	263	251	250	261	263	266	272	(42)	
(43)		Edh at Noon	27	33	43	43	46	50	57	55	43	35	29	26	(43)	
(44)	All-Sky Solar Radiation	RadAvg	969	1239	1605	1957	2199	2389	2399	2171	1787	1342	1020	888	(44)	
(45)		RadStd	63	108	107	97	121	94	51	60	72	86	70	57	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	+1.64	N/A	N/A	N/A	N/A	N/A	+196	(46)
(47)	Regional (4 neighbors)	N/A	N/A	-1.55	+1.67	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page